FeathersJS MCP Server MVP Demo

FeathersJS

Team C

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Original Vision & The Problem

Original Design Plan

- `search_docs`
- `get_template`
- `generate_service`
- `validate_code`
- `troubleshoot_error`
- 15 MCP tools total

What Peer Testing Revealed

- AI **already** formats/explains code
- validation/template tools = what Agents do anyways
- Root need: high-quality context, not output tools

What We Built Instead

Dropped 13 tools. Kept the core insight

`search_docs` (Kept + Upgraded)

- BM25 → BGE-M3
- Source-deduplication
- Token-budget trimming
- Version filtering
- Fully offline

Contributor Pipeline (New)

- `submit_documentation`
- `update_documentation`
- `remove_documentation`
- 6-stage validation
- Network-tier gate

MVP Scope and Traceability

In Scope

- Search_docs
- v5 + v6 knowledge base
- `submit_documentation`
- `update_documentation`
- `remove_documentation`
- 6-stage validation pipeline
- Rate limiting
- Network-tier gate
- Offline fallback

Out of Scope

- `get_template`
- `validate_code`
- `troubleshoot_error`
- Auto-updating docs from releases
- Web-based UI for contributors
- Multi-instance rate limiting

Dropped

- BM25
- All 13 original tools
- TypeScript AST
- ESLint/Prettier
- Template fragment composition

Requirements Coverage

1 FR-006 / US-001 — Docs search

2 FR-Search — Version filtering

3 NFR-001 — 2s response time

4 NFR-003 — Setup ≤ 3 commands

5 Contrib — Submit new docs

6 Contrib — Update existing docs

7 Contrib — Remove docs

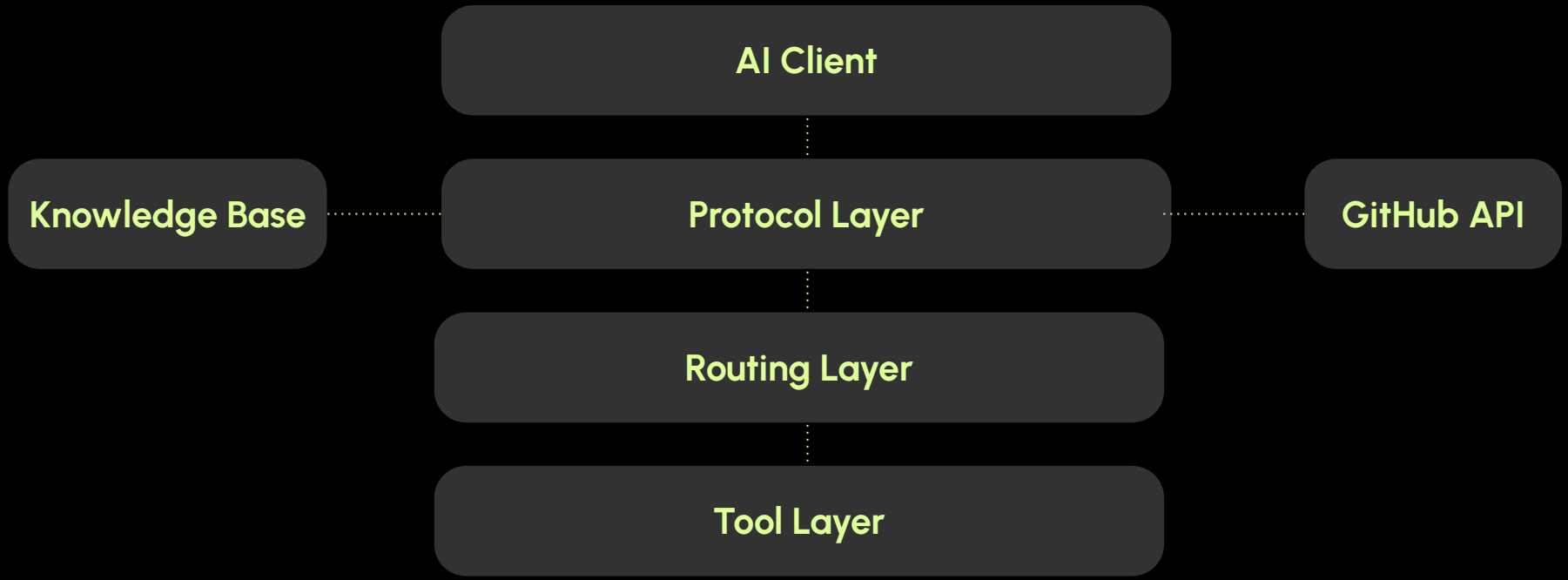
8 Security — Input validation

9 NFR — Network isolation

10 NFR-004 — Code quality (80% cov)

11 FR-003 / US-002 — Code gen (dropped)

System Architecture (Revised)



Live Demo

User Story → Workflow → Result

Scenario Setup 1

Ava is building a FeathersJS v6 service

`search_docs` 2

Query: "how do around hooks work in v6?"

Contributor Flow 3

Marco notices a gap in docs. Calls `submit_documentation`

6-Stage Validation Pipeline

Schema 1

Required fields, types, lengths (AJV)

Path Restriction 2

Regex allowlist, no .. / null bytes / backslashes, version–path match

Content Sanitization 3

Rejects <script>, <iframe>, javascript: URIs,
large data: URIs

Markdown Lint 4

Requires # Heading, minimum prose content

Existence Check 5

submit: warns if exists.
update/remove:
rejects if missing

Rate Limiting 6

1 operation per 60 seconds per server instance

Reflection: What We Learned

1

AI Agents already do part of what we planned

2

BM25 vs. BGE-M3 — quality gap was real

3

Peer testing saved ~40 hours of wasted work

Plans & Risks

Remaining Features

Hybrid search fallback	HIGH
Admin review workflow	HIGH
Knowledge base updation	MEDIUM
Expanded v5 doc coverage	MEDIUM
Performance profiling	MEDIUM
Platform guidance	LOW
Multi-instance rate limit	LOW

Risks & Blockers

Knowledge Staleness	HIGH
GitHub token scope errors	HIGH
BGE-M3 rebuild time	MEDIUM
MCP SDK breaking changes	MEDIUM
Rate limit too aggressive	LOW

Team Contributions

1

Protocol & Routing

2

Search & Embeddings

3

Contributor Pipeline

4

Testing & QA

5

Knowledge Base & Docs

Thank you

Questions?